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Way of the DAO

ShapeShift as a Decentralized Organization

> ShapeShift is but one of many lifeforms beginning to walk upon this new fertile plane, and we're finding the landscape quite appealing.

When I first heard of DAOs I was skeptical—I dismissed them as hippy collectives; hallucinations of the rainbow unicorn ETH folks.

But ShapeShift is 10+ months into our DAO experiment, and at this point I'm a firm supporter of the concept. We've progressed from "interesting theory" to actual execution.

DAOs as an organizational structure probably aren't appropriate for many orgs, but they are a powerful social innovation worthy of study and experimentation. It's apparent that a new model of human economic coordination has been unlocked.

As perhaps the first traditionally-structured VC-backed corp to transition fully into a DAO, here's an overview of ShapeShift's prior form as a centralized entity, and how it now compares as a DAO.

ShapeShift's Structure as a Corporate Entity

ShapeShift was structured as a Swiss AG corp (roughly equivalent to a C Corp in the US) since 2015.

ShapeShift AG had a few subsidiaries, one in the US for employment of the team (most employees were in the US, so this was simpler than hiring everyone from a Swiss entity). The other subsidiary was a UK corp: the operating entity that interacted with customers (Terms of Service was between the user and this UK entity). There was also an Israeli entity from our 2020 acquisition of Portis, and the Swiss parent further owned equity stakes in a few crypto startups and a Swiss bank.

ShapeShift's Swiss entity had typical investors: a few dozen parties across Angel (2014), Seed (2016), and Series A (2017) investment rounds, and the latter round included several VCs such as Earlybird and Lakestar in Germany, Access Venture Partners in Colorado, FundersClub in San Francisco, and crypto-native Blockchain Capital, Pantera, and Digital Currency Group. Our Series A of ~$10m closed in March 2017, and we hadn't needed to raise money since then.

We had at times been profitable and generally held a large portion of our assets in real money (i.e., BTC and ETH) rather than fiat, so our balance sheet grew over the years despite a particularly brutal 2018-2019 period of losses.

ShapeShift AG had a typical Board that consisted of 4-5 members, always including me as CEO plus a seed-round representative and partners from two of the VCs.

Office and Human Resources

Our first office in 2015 was a 1,600 sq. ft. basement suite in the Capitol Hill area of Denver, and by late 2017, we'd moved into a beautiful, whole-floor 12,000 sq. ft. office downtown, costing around $35k/mo. plus a half million dollar initial build out.

Also by 2017, we had constructed a robust HR department, employing up to half a dozen people and several SaaS tools. Employees were usually traditional W2s, hired by the US entity, and we had employment contracts drafted by HR lawyers at great expense. Payroll was done through Gusto and payments happened every two weeks. Payroll

We implemented policies, of course. Lots of policies. Why? Because companies have policies and the experts will stress the importance of these policies. Vacation policy. Dog policy. Side-gig policy. Parking policy. Crypto trading policy. Harassment policy. Security policy. Expense policy. We probably had a Policy policy. All of these took time and money to create, update, and enforce.

Corporate ShapeShift offered health insurance to employees, because that's also just what companies do (at least since WWII when the practice began as a way to avoid government price controls). Company time went into assessing healthcare plans each year, and trying to squeeze dozens of individuals into a few options. It always felt weird that a crypto exchange was managing peoples' health insurance plans.

Anyway, all of this is quite typical for corporations. We engaged in these efforts because professionals advised us to. It's worth asking whether any of them helped our customers, or improved our technology.

Team Structure

I was CEO, serving at the discretion of the Board. I had an Executive Team of 4-6 individuals, such as a CFO, COO, CMO, Chief of Staff, Chief of Legal, etc. ShapeShift had several standard departments such as Marketing, Engineering, Product, Customer Support, Legal, HR, Operations, etc., and each department had a specific leader.

As we matured, we got into a routine budgeting process. Each year, and each quarter, expenses were reviewed and department budgets established. The control over these budgets ultimately rested at the Board level, which had to approve the annual spend.

And also as we matured, our Legal department grew. Most startups might have a retainer with outside counsel, or, if they need lots of guidance, maybe they have one in-house lawyer. By 2018, ShapeShift's legal team was 6-8 people plus multiple outside firms in multiple countries (we had to analyze matters under Swiss law, US law, and UK law… and if we desired to hire someone we'd have to analyze the HR law of that country as well; consequently it became policy never to hire anyone in France).

At its peak, ShapeShift spent over a million dollars *per month* on Legal. Even writing that now makes me throw up a little.

Overhead Abounds

All these rituals mentioned above are just a description of overhead: the costs borne by ShapeShift *merely to administer and operate the corporate entity structure.* Notice there has been zero discussion yet of what we did as a business: building a non-custodial crypto exchange platform.

and regulatory optimization. These resources were spent on the following questions:

- How many entities do we need, where should they be based, and what form should they take?
- How do we file appropriately, who should do the filing, and when should the filing be filed?
- Have the files been audited? Were the audits approved by the right agency at the right time?
- Is a given policy applicable in this jurisdiction, or do we need to redraft a new one with separate local counsel?
- If an entity is structured in one way, how does that change our legal analysis in other jurisdictions or with other entities?

Most of this cost (time, money, attention) supports the maintenance of the legal industrial complex itself; it does not directly support our customers nor develop our product.

The modern corporate entity endures under a preposterous amount of overhead and associated bureaucracy. Much of this is dead-weight loss, borne of tradition, insidious legal incentives, and a lack of alternatives.

Okay, So How Does the ShapeShift DAO Differ?

First, let's remember that DAOs are completely experimental, and "best practices" are not yet established. DAOs can take many forms. They can be small or large. They can have a narrow purpose or broad. They can seek profits or pursue philanthropy (or why not both?).

ShapeShift's DAO starts with a specific purpose: to build an open-source crypto platform for the world. This platform must be non-custodial, and multi-chain. It cannot require KYC.

The number of products that fit that description today is shockingly small. We think it's crucial, and the ShapeShift DAO exists to see this vision realized.

The platform can and does generate revenues, but there is no explicit mandate to maximize returns for any set of stakeholders. We are long-term focused, with no ability or desire to ever "go public," a Faustian bargain for any crypto firm that holds user-sovereignty as a core principle.

ShapeShift's DAO-built crypto platform (now open-sourced) launched in the fall of 2021, a few months after we announced our intention to decentralize. This new platform is neither owned nor controlled by ShapeShift AG. It's a public good owned by humanity, and controlled by the aggregate holders of FOX tokens. Try it out at app.shapeshift.com

To fulfill its purpose (to build an open-source crypto platform for the world) ShapeShift needs resources.

The ShapeShift DAO today controls a pile of resources locked in a Treasury (specifically a Gnosis Safe multisig wallet). The DAO Treasury has over $50 million of assets, and from this, all economic activity of the DAO happens. The balance comprises ~85% FOX Tokens, plus USDC, ETH, and several DeFi assets.

Note what we don't have as a DAO: a bank account. ShapeShift AG (the corp entity) maintained multiple bank accounts in multiple countries. We were always at risk of them being shut down, and several were closed over the years, causing disruption and anxiety. All the administration, bank manager "relationships," and risks surrounding those custodians is now eliminated from our operations.

Friendly advice for all would-be DAOs: having a bank account is a non-starter. Banks require entities. Entities are antithetical to decentralization.

If the DAO wants to be conservative, it can allocate more assets to stablecoins. If it wants to earn interest or get a loan, DeFi already provides better options than a bank.

The security is superior too: cryptographic signatures to move funds vs scribbled "signatures" on paper or by speaking social security numbers over the phone.

Each of our bank accounts took weeks to set up. A DeFi account takes 5 minutes.

As a DAO, we've truly left the banking world behind and it feels great.

Controls, Governance, Structure, and Funding

The ShapeShift DAO has no legal entity, no CEO, no Board, and no shareholders.

Instead, it has token holders, in which all control resides.

There are around 35,000 holders of the FOX Token today, all pseudonymous, from all over the world. FOX has a fully diluted market cap of around $80m and is traded on several major exchanges including Uniswap and Coinbase. There are 1,000,001,337 FOX Tokens. No more will be created (no inflation).

Neither ShapeShift nor myself have ever sold any of these tokens. They were distributed in the world's largest airdrop last summer to over a million potential addresses, which included ShapeShift users, users of other noble DeFi projects, our beloved employees, and our shareholders (more details here).

FOX holders wield all control, and can subsequently form any organizational structures and fund any initiatives they find desirable.

The ShapeShift DAO uses a combination of Snapshot and Boardroom for voting (tooling for DAOs is an obvious new growth area for software). Voters cryptographically sign ballots with their Ethereum wallet. Because these signed messages are stored off-chain on IPFS, voting doesn't cost any gas. Each Governance proposal specifies a certain block height, where the FOX balances are tallied (a good practice is to set the block height in the past to prevent gaming).

Let's look at a hypothetical example of how this works:

- ShapeShift (as an app) needs Marketing (like any startup)
- Alice wants to work for a DAO, and is a skilled marketing leader.
- Alice posts on the ShapeShift Forum, conveying who she is, why she is credible, and what she proposes.
- Alice proposes creating a Marketing "Workstream" (our name for a department). She wants $100,000 in FOX or USDC paid monthly to Address X over six months. With this money, she'll recruit a team and do a great job marketing ShapeShift.
- People discuss the proposal in the forum. Upon apparent community support, Alice drafts a formally structured Proposal, which she posts on Snapshot.
- After a week, Alice's Proposal hit quorum and passed with 90% approval of FOX voters.
- The Treasury now sends $100k to Alice each month for the six month term.
- After six months, Alice will either a) not renew or b) ask to renew a similar agreement. If she has convinced the community she's done a good job, the community can vote again for a new agreement.

Permutations of this process are how any DAO makes things happen.

Note that proposals can be big or small, but there are procedural guardrails in place. Perhaps Bob is a master of SEO and wants $5k to engage in a campaign. This could be done by a formal vote, but is too small of an activity to warrant formal voting, and he'd be told as much in the forum discussion.

It would be better for Bob to go to Alice and convince her that he's worthy of contributing to the marketing effort, and Alice could fund him with $5k from her large marketing budget. Whether the DAO has any knowledge of this sub-contract is up to Alice and Bob.

In this way, granular expenses don't clog up the voting/governance process (but a small one-off vote always **could** happen if the community desired it).

Consider also that under this structure a workstream leader (Alice) can utilize whatever discretion she wants in the establishment of her team and strategy. She will probably need to communicate what her plans are in order to be funded in the first place, but any of the following could happen:

- Alice could already have a small team she works with. That team might even be a company (Alice's Marketing Guru's LLC). She may disclose who is on the team or not. She may disclose how they're paid or not.
- Alice could build a team after the Workstream is approved/funded. How she builds such a team is up to her.
- Alice could work with people in any country, at any price. She cares not about the HR regulations of France and can thus hire Pierre, the analytics guru that Corporate ShapeShift could never hire due to policy.
- Alice could have no marketing experience at all, but instead simply sources a kickass firm to deliver the results. She takes $20k off the top per month and the DAO gets great results from that firm.
- Alice might be mediocre and her workstream won't be renewed next time.
- Alice might be terrible and the community could vote to terminate the workstream prematurely (ending the $100k payments).
- Alice may be doing okay, but Charlie knows he can do better. He proposes a competing Marketing workstream and might go head to head against Alice. Have you ever heard of a company with two marketing departments? Wouldn't that be interesting to try? DAOs open up a world of dynamic competition **within** an organization.

Once Alice receives funding for her newly created Workstream, she can decide how to pay contributors. In ShapeShift, most Workstreams are utilizing a platform called Colony to manage payments to their members as it's pseudonymous and transparent (and uses an Ethereum side chain to save on tx fees).

Consider here the difference between Corporate ShapeShift and DAO ShapeShift for payroll:

- Corp ShapeShift would pay for an expensive SaaS product (perhaps with a lengthy contract term) to manage payroll. This product would inevitably only work in some territories (often only the US), would require banks, and take days for processing. A bunch of money would be stolen by the government ("payroll taxes").
- DAO ShapeShift uses open source apps that cost nothing and work everywhere, with no bank involvement and instant payment with mathematically provable, real-time auditability. No payroll tax money would be stolen by the government because there are no W2 employees (certainly income taxes are still paid by individuals, but that's also the case in corporate form).

The ShapeShift DAO has no set number of workstreams or leaders. It can adapt to market conditions. Crypto winter? It can shrink down to a few dedicated community members to keep the flame alive. Boom time? It can potentially grow faster than a centralized org given the lack of HR friction.

Anyone in the community that sees a need can propose something, and a vote will make it so or not.

ShapeShift has had around 80 proposals in the 10+ months of DAO-hood, and nearly all of them pass. Why? Because we have established a norm by which matters generally don't go up for formal vote until there is clear community support. Anyone could put up a vote on a whim (thereby violating the norm) but the community would likely kill it purely on procedural grounds. Some Proposals have been pulled by the proposer when the vote was too contentious. In multiple instances, adapted versions of removed proposals that sufficiently address or incorporate community feedback have been passed.

This kind of decentralized order is what the crypto revolution is all about. The DAO conforms to code, game theory and economic incentives, and shared community norms. The reason the concept works is because it serves both individual and collective interest through iterative self-selection.

Anyone who doesn't believe they are personally served by the ShapeShift DAO can exit and sell their tokens. Those interested in our mission can participate and acquire tokens. This process happens in real-time, 24/7, rather than in "investment rounds" and "liquidity events" with years devoid of liquidity in between.

Thus, the community at any given time is almost by definition the set of individuals who a) like our vision and b) believe they can benefit from it and/or contribute to it as individuals.

And because anyone in the DAO is empowered to speak their mind on the public forum or Discord channel, as a DAO, ShapeShift enables a true "voice and exit" principle that no corporation can match.

Coordination and Communication

As a typical corporation, ShapeShift had a physical office where >90% of employees worked in-person. Internal email was occasional, but Slack was ubiquitous.

We left the office in March 2020 in anticipation of the COVID lockdowns and found that we were quite effective as a remote company. This was one of the prerequisites to DAO-hood for us: a fully-remote workforce.

As a DAO, communication now happens in several ways:

- Real-time, high-frequency chat occurs in our Discord
- We have a schedule of DAO meetings each week, executed as Discord voice chats.
- Specific Workstreams (i.e., departments) can create their own policies for communication (e.g. a private daily standup and public weekly office hour)
- More formal, structured discussion exists as threads in our Forum
- Actual governance proposals follow a template and are posted on Snapshot.

- In-person events occur periodically. The DAO threw a notorious raging blizzard party to kick off ETHDenver (funds were raised for catering, bar staff, transportation, and cleaning). The Engineering and Product Workstreams have held in-person offsites focused on intensive collaboration.

- ShapeShift communicates out to the public and our users through the normal social media channels (mainly our Twitter and YouTube), email, and Discord. We are branching into new Web3 communications channels as they develop.

Importantly: Organizational communication has become public. Our users hang out in the same Discord channels where we discuss our product and marketing strategies. We post video recordings of all scheduled meetings (Netflix famously publishes video recordings internally… we've gone one step further). From an engineering perspective, ShapeShift is open-source now, so our code, submitted pull requests, and the discussion between engineers is all public.

However, we have no policy mandating public communications. Anyone within the DAO is free to have private discussions with and among each other if they wish. Indeed, it'd be impossible to enforce anything else.

It's worth pointing out that in the "regulated" world of corporate America, sincere transparency is rarely found. All the regulations put forth by the SEC, for example, trying to mandate "disclosures" and "accountability" pale in comparison to what ShapeShift is doing voluntarily as an unregulated, free-market, open-source DAO.

But Is It Decentralized?

Our industry struggles with the concept of decentralization, and rightly so, for it's the core innovation and most important consequence of Bitcoin's genesis.

Decentralization is clearly a spectrum (actually a multivariate spectrum). There exist numerous variables of any crypto project, which can be framed as more decentralized or less decentralized.

We can acknowledge for example that Ethereum is more decentralized than Bitcoin in terms of application developers. However, we must also recognize that Bitcoin is more decentralized than Ethereum in terms of node topography and infrastructure.

The variables can be compared internally as well. Bitcoin is more decentralized in its user base than in the makeup of its miners, and Bitcoin as a whole is more decentralized today than it was 10 years ago. Hopefully it is less decentralized today than it will be 10 years from now.

Crypto projects all move along these multivariate decentralization spectrums, and DAOs are no different.

ShapeShift as a DAO today is far more decentralized than it was as a corporation. If it's designed correctly, with time it will further decentralize upon a broad and diverse community of participants.

But I am not the CEO of the DAO, neither in title nor practice. In fact, I have no formal role whatsoever. I'm not leading any of the Workstreams. I'm not managing anyone. I am not even a signer on the multisig treasury. If someone has an idea, they don't need my approval. If someone wants to hire another, I'm not required in any part of that process. Similarly, I answer to nobody. I required no approval for this blog post; my words don't represent the DAO in any official capacity. And yet, I voluntarily sought feedback and edits, as I'm part of a community that I value.

But who's in charge?! Literally no specific person. Specific individuals are in charge of specific Workstreams or projects, but they hold no eternal title to them and they—or the workstream itself—can be replaced upon a vote.

But the DAO is not a squishy Kumbaya circle of equality, either. It has structure, order, and hierarchy. The structure emerges organically from the desires of token holders. 1 FOX = 1 vote. And because the token is trading 24/7, control over this organization is changing minute by minute across tens of thousands of pseudonymous individuals.

Over time, control will tend to flow from those who lose interest in it to those who find interest in it. The org can thus travel across time and space, detached from any jurisdiction and beholden to no specific person; it exists as a buoyant Schelling Point around a specific vision, guided by the economically weighted interests of its community.

ShapeShift was able to decentralize once it no longer needed anything from the analog world—no bank account, no office, no licenses or arbitrary permission slips—and as such, it's able to transcend a great many obstacles which, in its prior form, were existential.

DAOs Have Emerged

Nothing in this post should lead readers to the conclusion that DAOs have no problems.

They are certainly not utopian. They don't magically solve the inherent conflicts between individuals working together toward a common purpose. While they eliminate a great many friction points from traditional corporations, they also introduce new issues and different friction points.

Corporate structure is a social discipline which has been refined over centuries, through trial and error, and the reflections of a great many brilliant minds. Much has been learned about the right ways to build companies, and much needs to be learned about the right ways to build DAOs. Entrepreneurs and organizational thinkers should be thrilled by the new design space.

DAOs may be most potent at the intersection of a) digital platforms and b) heavily regulated industries. This is precisely where crypto sits, and this is why, unsurprisingly, DAOs have emerged as a new organizational species.

ShapeShift is but one of many lifeforms beginning to walk upon this new fertile plane, and we're finding the

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Erik Voorhees

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and we may now rejoice
at the next century of its
consequences.